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Subject: Azure Fundamentals

Unit 3 : Security, Privacy, Compliance, and Trust:

Computer Science & Engineering

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Outline

- Introduce the Azure security features
- Identity and access management
- Azure Active Directory and data Protection
- Compliance frameworks
- Azure governance methodologies

Introduce the Azure security features

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Azure offers a comprehensive suite of security features to protect data, applications, and infrastructure in the cloud. Here's an overview of key Azure security features:

1. Identity and Access Management (IAM)

- **Azure Active Directory (Azure AD):** Centralized identity management service that provides single sign-on (SSO), multi-factor authentication (MFA), and conditional access policies.
- **Role-Based Access Control (RBAC):** Enables you to assign specific permissions to users, groups, and applications at a granular level.

Introduce the Azure security features

2. Data Protection

- **Azure Key Vault:** Securely store and manage cryptographic keys, secrets, and certificates.
- **Encryption:** Data encryption at rest and in transit is supported using Azure Storage Service Encryption and Azure Disk Encryption.
- **Azure Information Protection (AIP):** Helps classify and protect documents and emails by applying labels.

Introduce the Azure security features

3. Threat Protection

- **Microsoft Defender for Cloud (formerly Azure Security Center):** Provides unified security management and advanced threat protection across Azure and on-premises resources.
- **Azure DDoS Protection:** Protects your applications from Distributed Denial of Service (DDoS) attacks.
- **Azure Sentinel:** A scalable, cloud-native security information and event management (SIEM) solution that provides intelligent security analytics.

Introduce the Azure security features

6. Compliance and Governance

- **Azure Policy:** Create, assign, and manage policies to enforce rules and ensure compliance with corporate standards.
- **Azure Blueprints:** Create reproducible, versioned sets of governance policies, control definitions, and other resources.

6. Monitoring and Logging

- **Azure Monitor:** Collects, analyzes, and acts on telemetry data from Azure and on-premises environments.
- **Azure Log Analytics:** Provides analytics and insights by querying and analyzing logs from various sources.

Introduce the Azure security features

7. Application Security

- **Azure Web Application Firewall (WAF):** Protects web applications from common vulnerabilities like SQL injection and cross-site scripting. It integrates with Azure Front Door and Application Gateway.
- **Azure App Service Security:** Offers built-in security features like managed certificates, HTTPS, and more for web applications hosted on Azure App Service.

8. Security Development Lifecycle (SDL)

Azure encourages adopting a Security Development Lifecycle, which integrates security best practices into the development process. This includes threat modeling, security design, and regular security reviews.

Identity and access management

Identity and access management

- Identity and Access Management (IAM) is a critical component of any cloud infrastructure, providing the necessary tools and processes to manage and secure access to resources.

- In Azure, IAM is primarily facilitated through Azure Active Directory (Azure AD). This tutorial will cover the key concepts, features, and best practices related to IAM in Azure.

Identity and access management

1. What is Azure Active Directory (Azure AD)?

Azure AD is Microsoft's cloud-based identity and access management service. It helps organizations manage user identities, access to applications, and security policies. Azure AD supports a wide range of use cases, including:

- **Single Sign-On (SSO):** Allows users to log in once and access multiple applications without needing to re-authenticate.
- **Multi-Factor Authentication (MFA):** Provides an additional layer of security by requiring two or more verification methods.
- **Conditional Access:** Enforces access controls based on specific conditions, such as user location or device compliance.

Identity and access management

2. Key Components of Azure AD

2.1. Users and Groups:

Users: Represents individual identities within the directory. Each user has a unique username and profile.

Groups: Collections of users. Groups can be used to manage permissions and access to resources efficiently.

2.2. Roles:

Role-Based Access Control (RBAC): Azure AD uses roles to manage permissions. Roles can be assigned to users, groups, or applications, granting specific permissions within the Azure environment. Key roles include Owner, Contributor, and Reader.

2.3. Applications:

Enterprise Applications: Applications integrated with Azure AD for identity management, SSO, and access control.

App Registrations: Represent applications that are developed and registered to be managed through Azure AD, providing APIs and other resources.

Identity and access management

3. Core Features of Azure AD

3.1. Single Sign-On (SSO):

SSO simplifies user experience by allowing them to access multiple applications with a single set of credentials. It reduces the number of times users need to log in and enhances security by reducing password fatigue.

3.2. Multi-Factor Authentication (MFA):

MFA enhances security by requiring two or more verification methods, such as a password and a phone notification. This helps protect against credential theft and unauthorized access.

Identity and access management

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Identity and access management

3.3. Conditional Access:

Conditional Access policies allow you to enforce specific access controls based on conditions such as user location, device compliance, or risk level. For example, you can require MFA when accessing resources from outside the corporate network.

3.4. Identity Protection:

Azure AD Identity Protection helps identify and manage risk events related to user identities. It provides risk-based conditional access policies and alerts administrators to potential security threats.

3.5. Self-Service Password Reset (SSPR):

Allows users to reset their passwords without administrator intervention. SSPR

Identity and access management

4. Implementing IAM in Azure

4.1. Setting Up Azure AD:

Create an Azure AD Tenant: This is the basic unit of Azure AD, representing a dedicated instance of the service.

Add Users and Groups: Create user accounts and groups as needed.

Assign Roles and Permissions: Use RBAC to assign roles to users and groups, granting them the necessary permissions.

4.2. Configuring SSO:

Integrate Applications: Register and configure enterprise applications or custom apps with Azure AD for SSO.

Manage Access: Define who can access specific applications using groups or individual

Identity and access management

4.3. Enforcing MFA:

1.Enable MFA for Users: Enforce MFA requirements for critical accounts or all users.

2.Conditional Access Policies: Use Conditional Access to enforce MFA based on specific conditions, such as accessing from a new location.

4.4. Setting Up Conditional Access:

1.Define Policies: Create Conditional Access policies to control access based on factors like device compliance, location, and risk level.

2.Monitor and Adjust: Regularly review and adjust policies based on changing security requirements.

Identity and access management

https://learn.microsoft.com/en-us/azure/security/fundamentals/overview

https://www.microsoft.com/en-in/security/business/security-101/what-is-identity-access-management-iam#:~:text=Identity%20and%20access%20management%20is,who%20need%20access%20have%20access.

https://www.youtube.com/watch?v=7SjmCmEQE0I

https://www.youtube.com/watch?v=JI_fvvekFwo

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